

Homework Assignment 1

Map making practice

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Instructions:

For this assignment, you will explore an environmental justice topic of your choosing. You should select a region, community, or environmental issue that matters to you.

- Create two maps that communicate an environmental justice issue
- Write a brief paragraph explaining what my map communicates.

I.Setup

```
# code wouldnt render without this second installation package
install.packages("sf")
```

```
install.packages("sf")
install.packages("stars")
install.packages("tmap")
```

I.II Upload Libraries

```
library(sf)
library(stars)
library(tmap)
library(here)
```

II.Upload data:

```
# read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
`/Users/amapolagarcia/Git/EDS223-
HW1/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
```

```
using driver `OpenFileGDB`  
Simple feature collection with 243021 features and 223 fields  
Geometry type: MULTIPOLYGON  
Dimension:      XY  
Bounding box:   xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350  
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# filter California  
california <- ejsscreen |>  
  dplyr::filter(ST_ABBREV == "CA")  
  
# filter to San Mateo County  
county <- ejsscreen |>  
  dplyr::filter(CNTY_NAME %in% c("San Mateo County"))  
  
# find the average values for all variables within counties  
california_counties <- aggregate(california, by = list(california$CNTY_NAME), FUN = mean)
```

III. Create two maps communicating an environmental justice issue

Mapping the percentage of low-income in San Mateo County, CA:

```
# Static map  
tmap_mode("plot")  
  
sm_lwinc <- tm_shape(county) + # defines the spatial object to plot  
  tm_polygons(  
    size = 0.8,  
    fill = "LOWINCPCT", #inputs variable from ejsscreen data  
    fill.scale = tm_scale(values = c("darkorange", "lightyellow")),  
    fill.legend = tm_legend(title = "% of Low-Income")  
  ) +  
    tm_title("Percentage of Low-Income in San Mateo County") +  
    tm_scalebar(position = c("bottom", "left")) +  
    tm_compass(position = c("bottom", "right"))
```

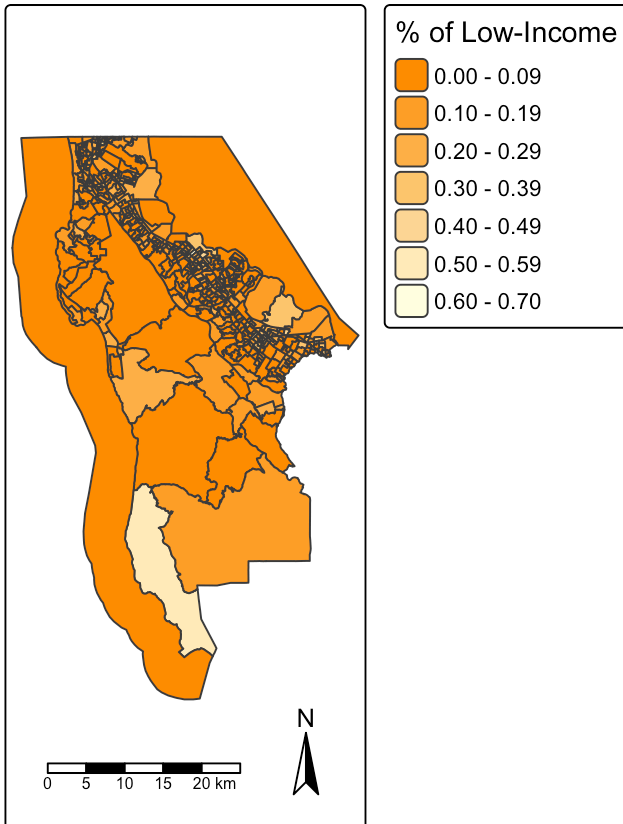
```
# Static map  
tmap_mode("plot")  
  
tx_air <- tm_shape(county) +  
  tm_polygons(  
    size = 0.8,  
    fill = "RSEI_AIR", #inputs variable from ejsscreen  
    fill.scale = tm_scale(values = c("-viridis")),  
    fill.legend = tm_legend(title = "Levels of Toxic Air Releases")  
  ) +  
    tm_title("Concentration of Toxic Air Releases in San Mateo County") +
```

```
tm_scalebar(position = c("bottom", "left"))+
tm_compass(position = c("bottom", "right"))
```

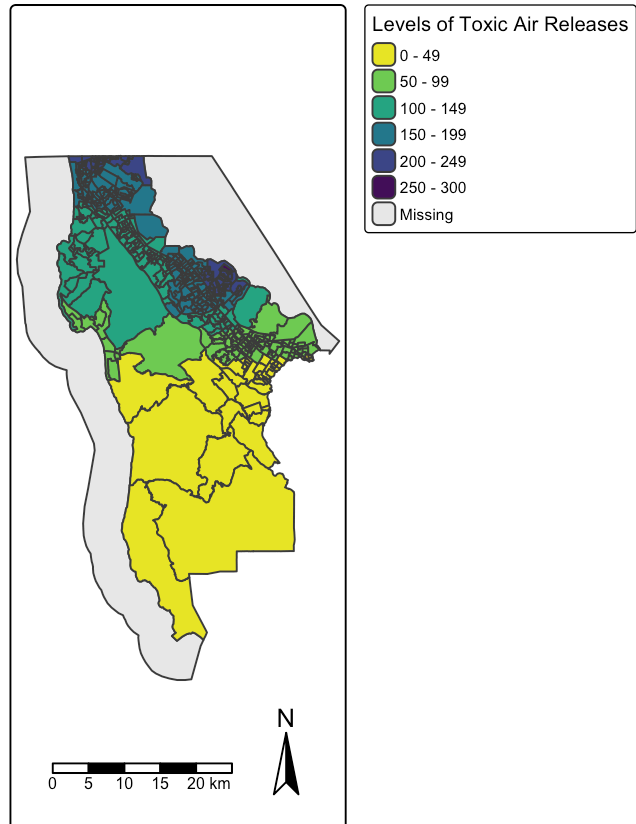
IV. Arrange both maps together

```
tmap_arrange(sm_lwinc, tx_air)
```

Percentage of Low-Income in San Mateo County



Concentration of Toxic Air Releases in San Mateo County



V. Analysis

When thinking about environmental justice issues, two variables from the ejSCREEN dataset were “% of low income” and “concentration of toxic air releases” within San Mateo County, CA. I wanted to see whether there was a relationship between the percentage of low-income communities and toxic air releases. I asked this because I lived in San Mateo for the last six years before moving to Santa Barbara. San Mateo County is known for being beautiful and green, with lots of nature nearby; however, where I lived, it was predominantly low-income, with industrial sites nearby and close proximity to freeways. Looking at both maps, I see that areas with a higher percentage of low-income people also have a darker shade, indicating higher levels of toxic releases in the same vicinity. I also see that in the second map, where toxic air concentrations are lower, the percentage of low-income people also drops. This suggests structural injustice affecting disadvantaged communities living in certain areas within the

county. My maps do have limitations, they show the county as a whole and combine all areas. In my future analysis, I would like to go deeper into the dataset by collecting more information on low-income percentages and showing what types of toxicity are polluting the air. I believe my maps show that in certain regions of the county with higher levels of low-income residents, there are also higher concentrations of toxic air releases.